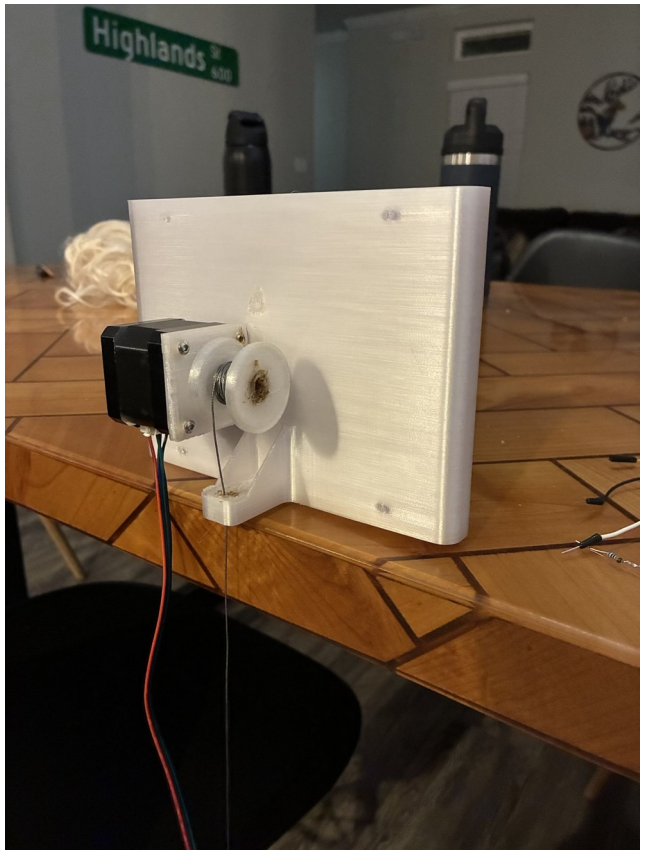
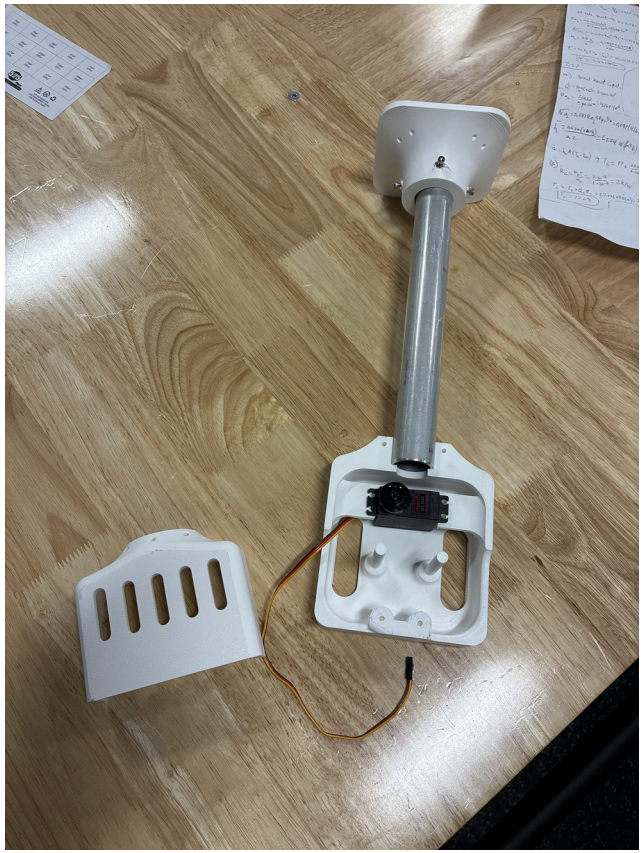
RoomCleaner

A four-cable ceiling robot designed to pick laundry off the floor

Jasper Buntinx
Mechanical Engineering, Texas A&M University
October 27, 2026



Wall station (built)

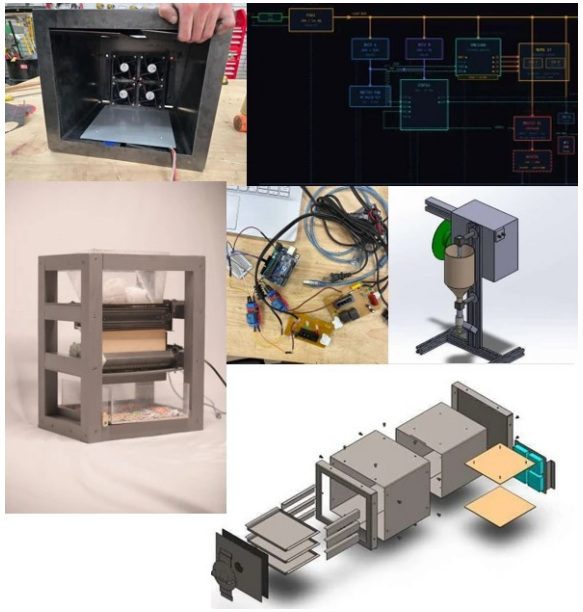


Claw body and servo (built)

About Me



Ironman 70.3, marathon



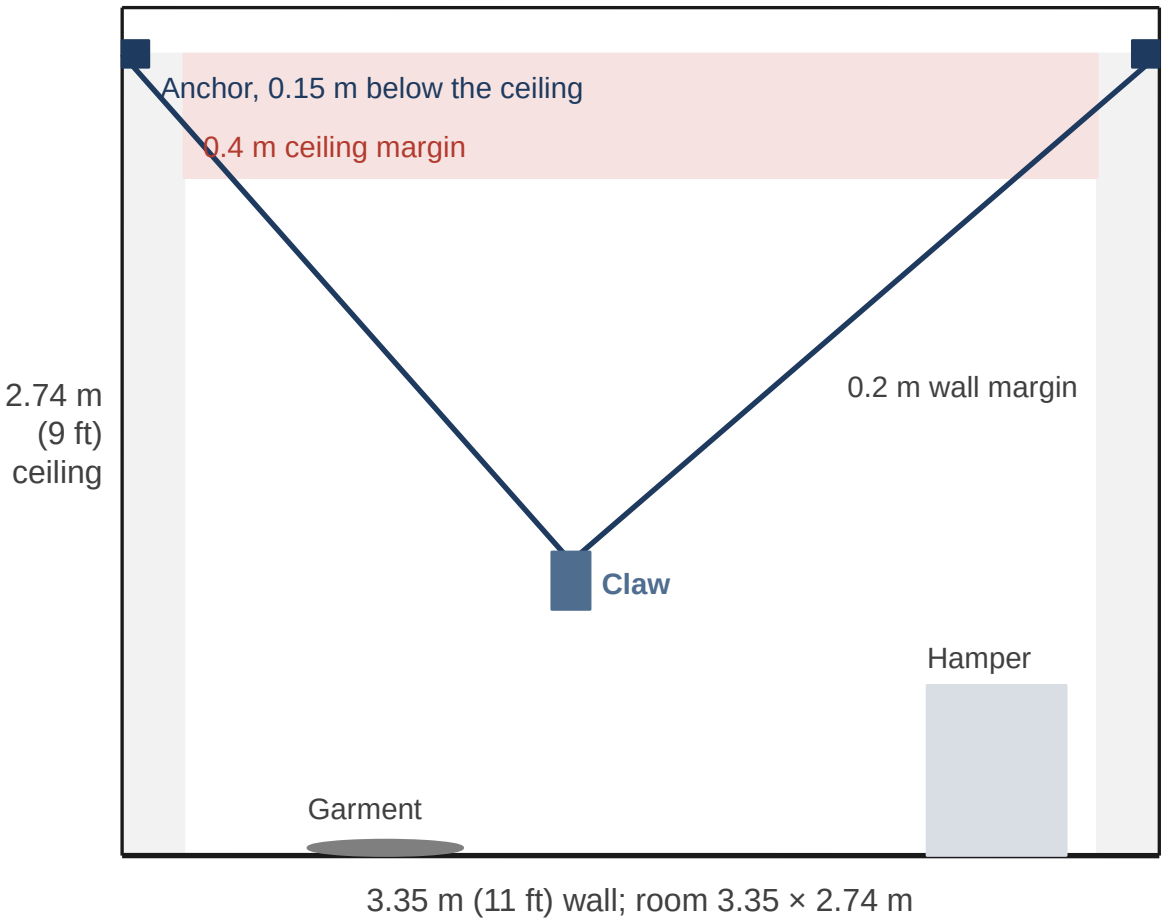
EcoFil filament recycler



Gladiator Dash obstacle build

- Mechanical Engineering, Texas A&M University, class of 2027
- Built EcoFil, a filament recycler with a shredder, dryer and extruder
- Build logistics chair, Gladiator Dash: obstacle assembly, repairs and transport ($185K raised)
- Ironman 70.3 in 6:53 and a marathon in 2025

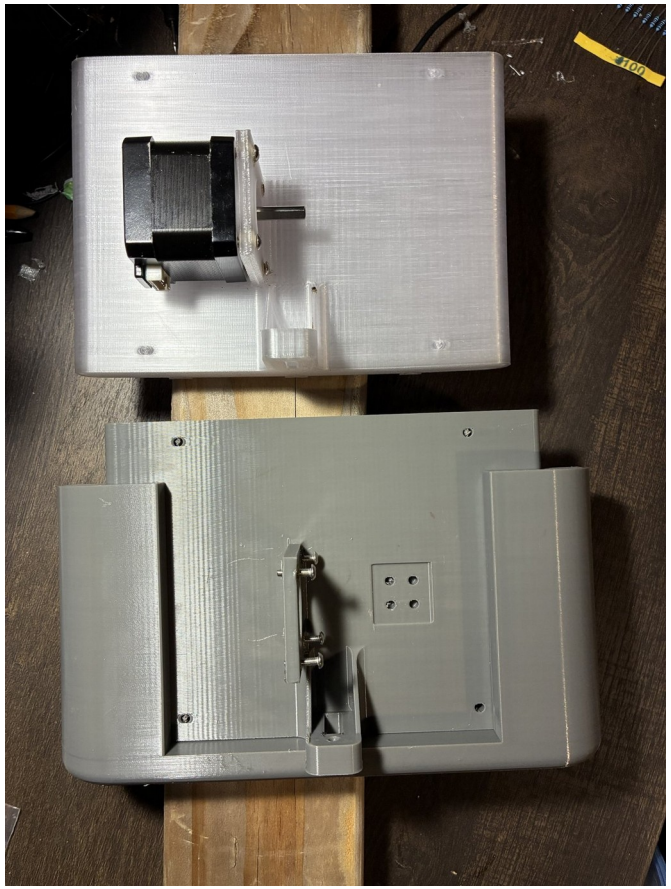
Project Goal

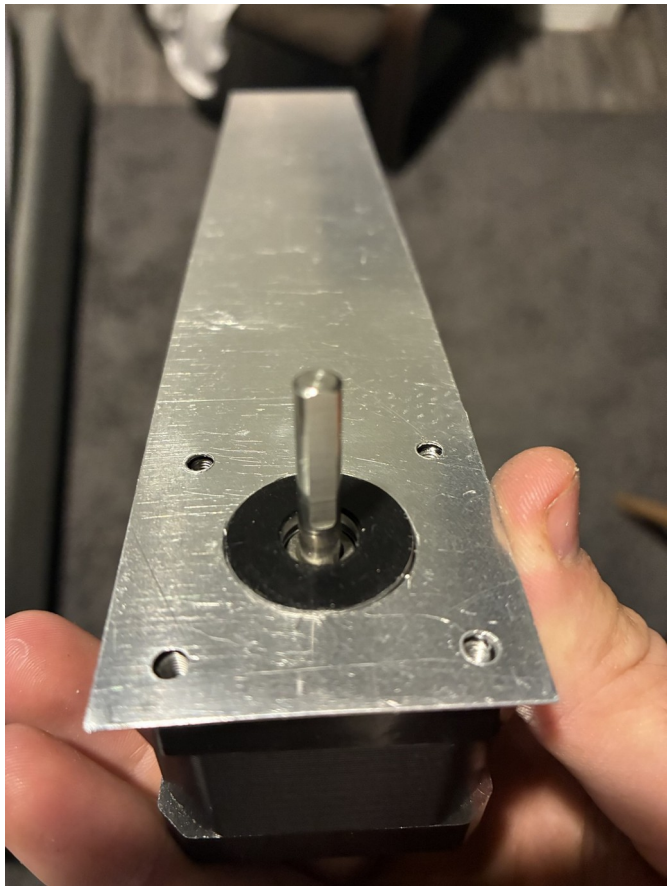


Room elevation to scale; seen from the side, the four cables appear as two

- Pick up laundry in a 3.35 × 2.74 m (11 × 9 ft) room under a 2.74 m (9 ft) ceiling, and drop it in a hamper
- Design load 0.9 kg (jeans); first test 0.45 kg (1 lb)
- Success: one garment from floor to hamper with no human touch; I approve each move on the phone
- Safe state: PAUSE holds position, and nobody stands under the robot

Where It Stands Today



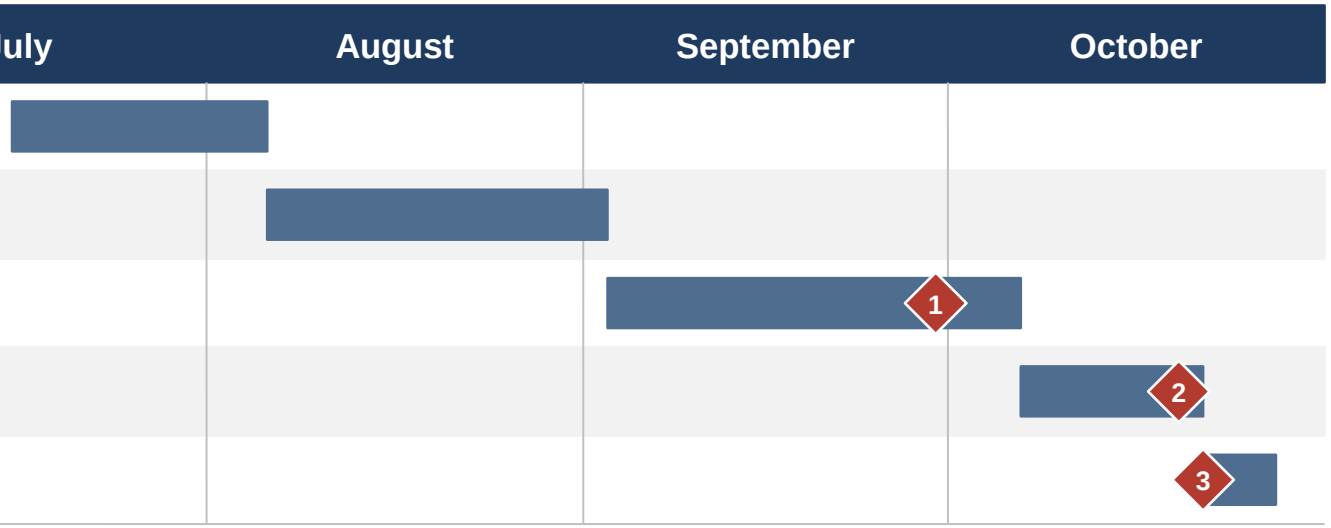


Built: station shells, and the prototype fin on its motor

- Built: *[how many]* wall stations, station electronics, the claw body and gear train
- Measured: claw 0.60 kg with battery; most other numbers in this talk are calculated or simulated, and labeled
- Open risks: lift margin rests on an unmeasured torque, jeans grip is short at narrow grabs, the prototype fin has 1–4 K of thermal margin
- This week: pull test, 90-minute hold test, three-station test

Timeline and Scope

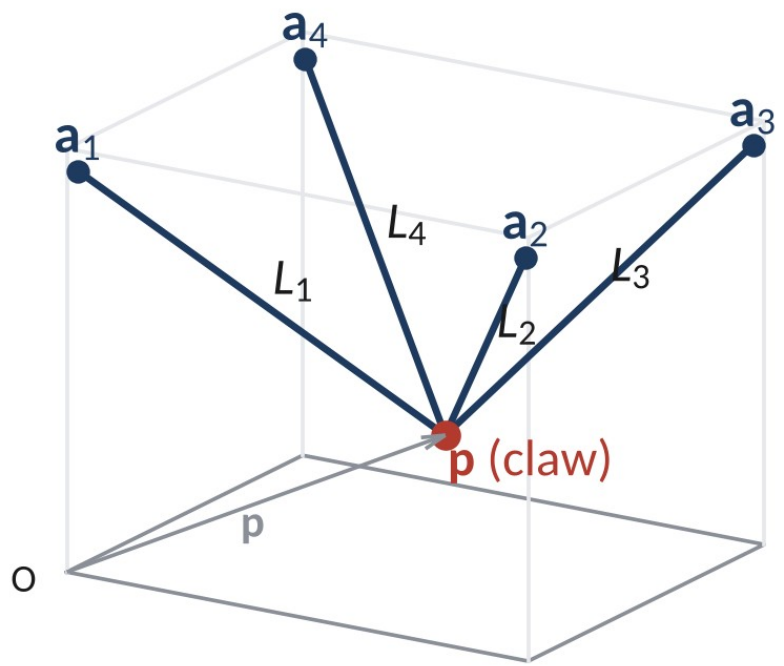
Phase	Dates
1. Concept	Jul 16 – Aug 5
2. Generation 1	Aug 6 – Sep 2
3. Redesign	Sep 3 – Oct 6
4. Final design	Oct 7 – Oct 21
5. Build and test	Oct 22 – Oct 27



1 Sep 30: first clamp's jaws cannot close on thin cloth 2 Oct 20: v6 clamp is 4.6 times its mass budget in CAD 3 Oct 22: drivers at 1.5 A; claw weighed

- Mine: requirements, design decisions, the SolidWorks station and claw, printing, machining, wiring and bring-up
- With Claude Code, under my direction: the control software, calculation scripts and CAD checks
- Purchased: steppers, drivers, controller boards, servo, camera, battery and encoder boards

Why a Cable Robot



	Floor robot	Gantry	Cable robot
Whole floor	Furniture blocks it	Yes	Yes
Floor left clear	No	Frame spans the room	Yes
Moving mass	Whole robot	Heavy beam	0.6 kg claw

- Chosen: four ceiling winches and one claw
- The cost: cables only pull, so cable tension sets the working space

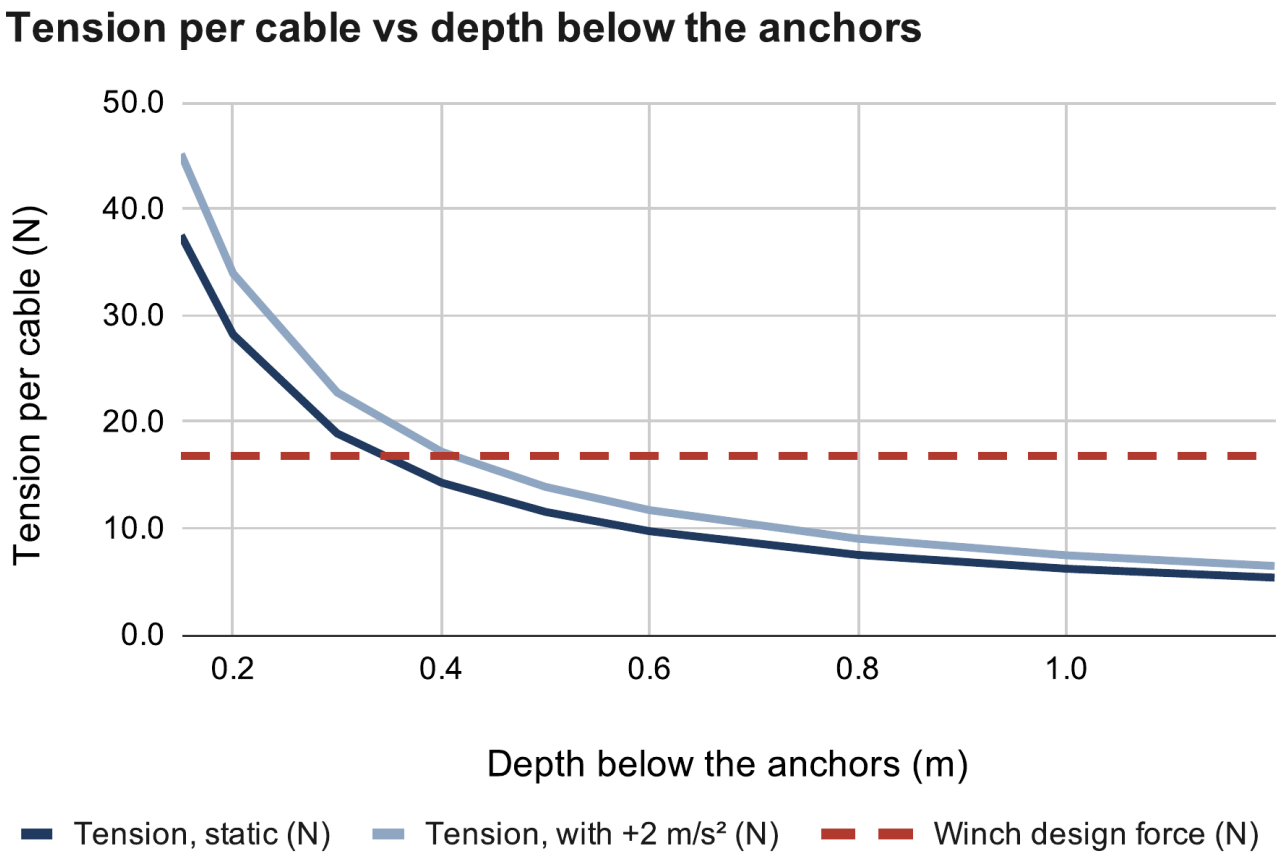
Cables from the ceiling anchors a_1 to a_4 to the claw at p

Design Trade-offs

Lever	Gain	Cost
Motor current up	Pull rises with current	Heat rises with current squared
Stronger claw	Stronger gears, firmer grip	More mass, less payload
Larger working area	More floor covered	Longer wires, more voltage drop
Closer to the ceiling	More working volume	Cable tension rises steeply
Smaller spool radius	More pull per amp	Tighter line bend, less room for the set screw
Larger safety margins	No skipped steps, no wall contact	Less usable force and floor

- The chain that binds: a heavier claw needs more pull, more pull needs more current, and heat rises with the square of the current

Cable Tension Sets the Ceiling Margin



Calculated (calc sheet: Cable tension)

- Weighed claw plus 1 lb, centred 0.4 m below the anchors: 14.3 N, 17.2 N at 2 m/s²
- That is at the 16.8 N winch limit; with jeans the claw must stay at least 0.6 m down
- The planner checks every cable at every path point against the measured pull limit

Tension per cable, 1 lb test at 0.4 m

$T = m\,(g + a) \,/\, (4 \sin \varphi)$

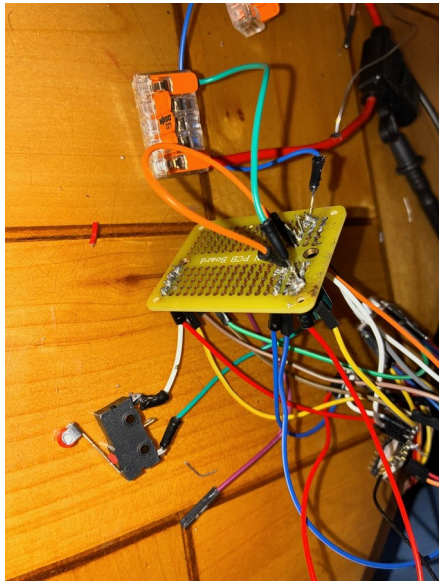
$= 1.054 \times 11.81 \,/\, (4 \times 0.181) =$ **17.2 N**

Winch limit: **16.8 N**

Generation 1: Bench Build and Findings



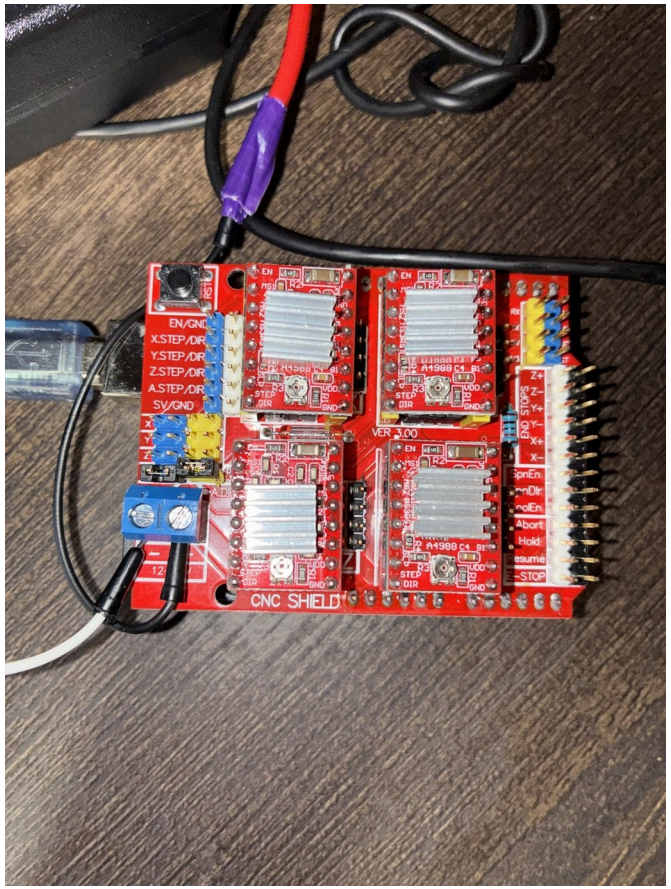


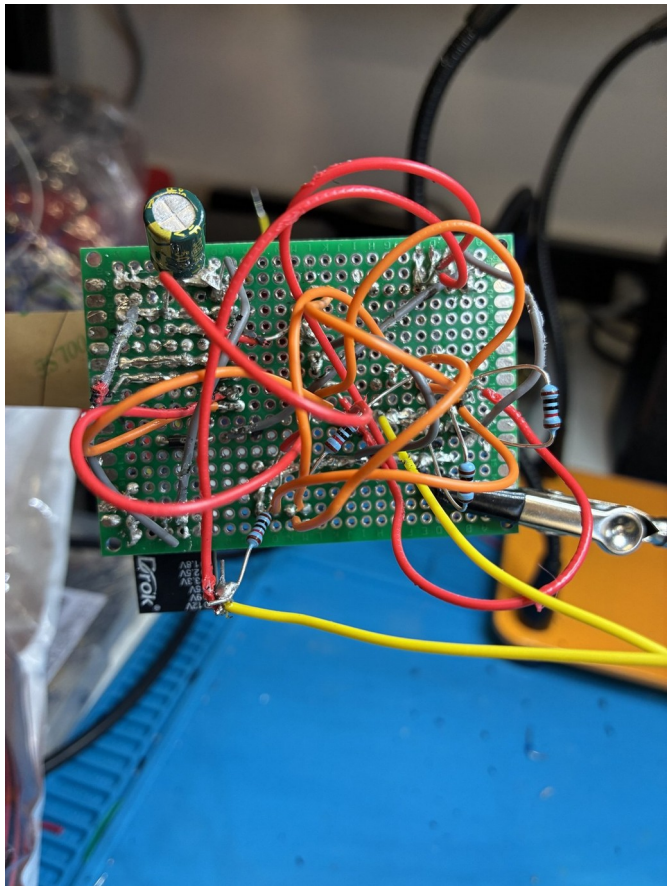


- A bench station lifted *[weight]*; the cable math held
- Bead homing would leave only 5–29 % of the room reachable, caught on paper before it was built
- One central board would need 10 wires to every corner once each winch had sensors
- Finger claw: designed in CAD (419 g of a 450 g budget), never built

Bench lift test, one-piece corner plate, homing switch

One Controller per Winch

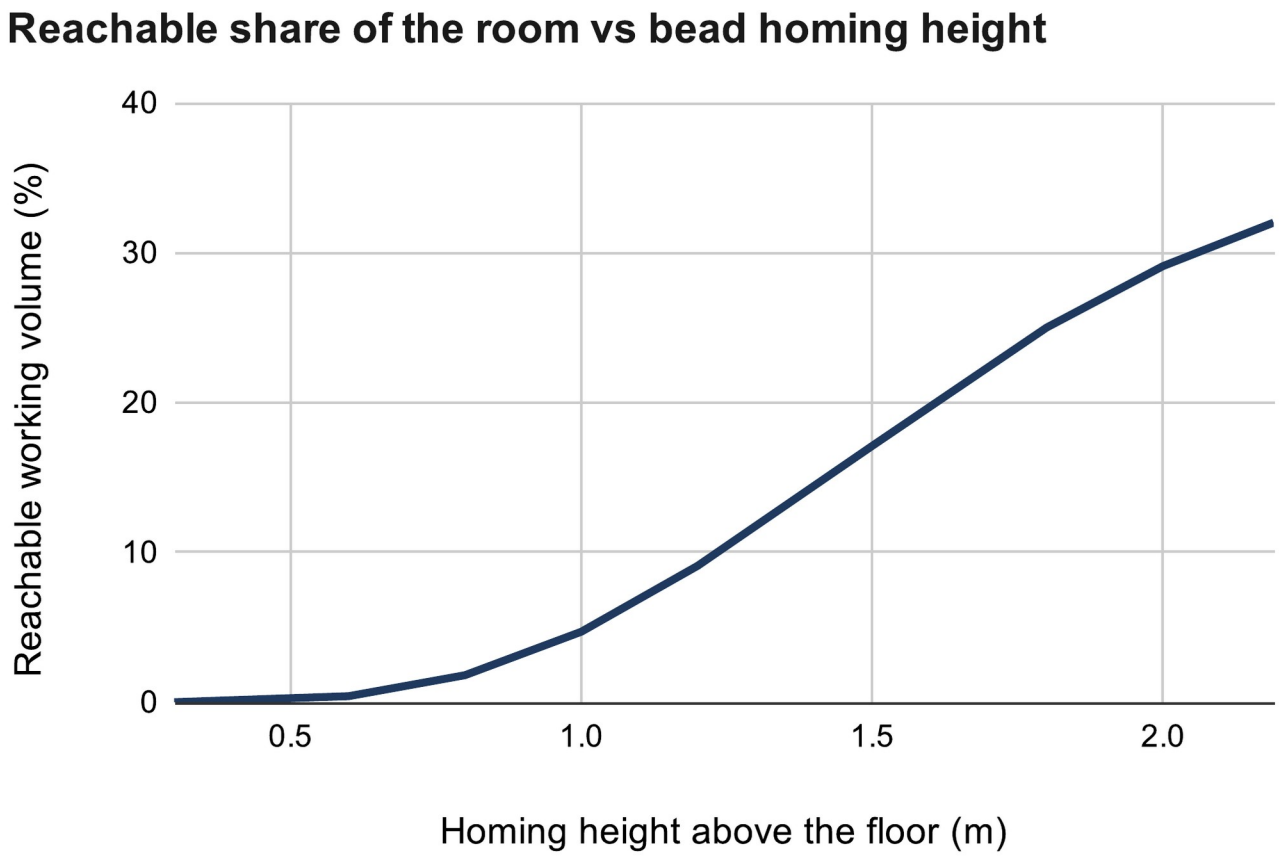




- Generation 1: one central board; with sensors it would need 250 m (820 ft) of wire
- Now: a controller in each station, fed by one 12 V pair: 50 m (164 ft)
- Each station can be built and tested on its own
- Cost: timing moves into software; 50 ms of skew is 2.5 mm of cable error, about 7.5 N

Before: one central driver board. After: a controller in each station.

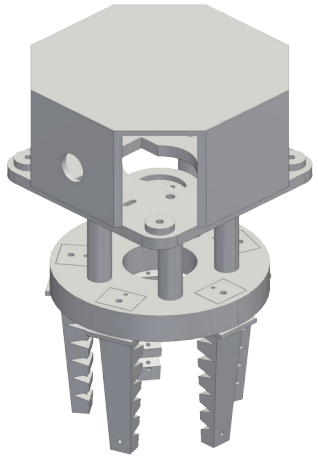
Camera Calibration Replaces Bead Homing



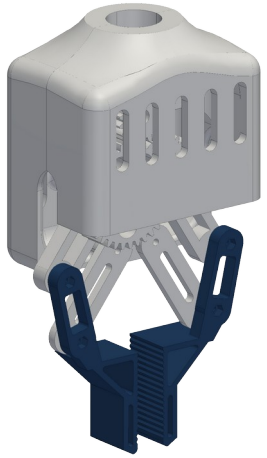
- Bead homing fixes each cable's shortest length, so most of the room is out of reach
- Instead the claw camera reads a printed floor board, and a fit finds the anchors and cable lengths
- Simulated: cable lengths known to about 12–16 mm, well inside the 0.2 m and 0.4 m margins
- The whole room opens only when the fit's error bars allow it

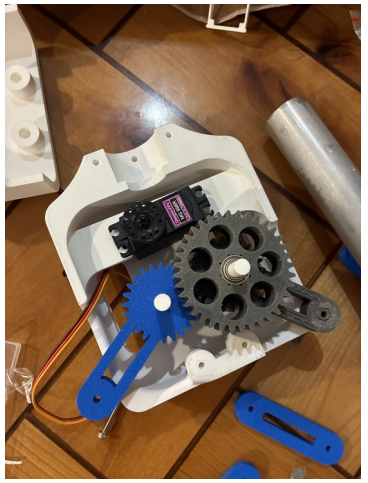
Calculated (calc sheet: Bead homing)

Claw: Four Versions to a Parallel Grip









Finger claw (CAD only), v6 clamp (CAD), final four-bar (CAD), printed gear train

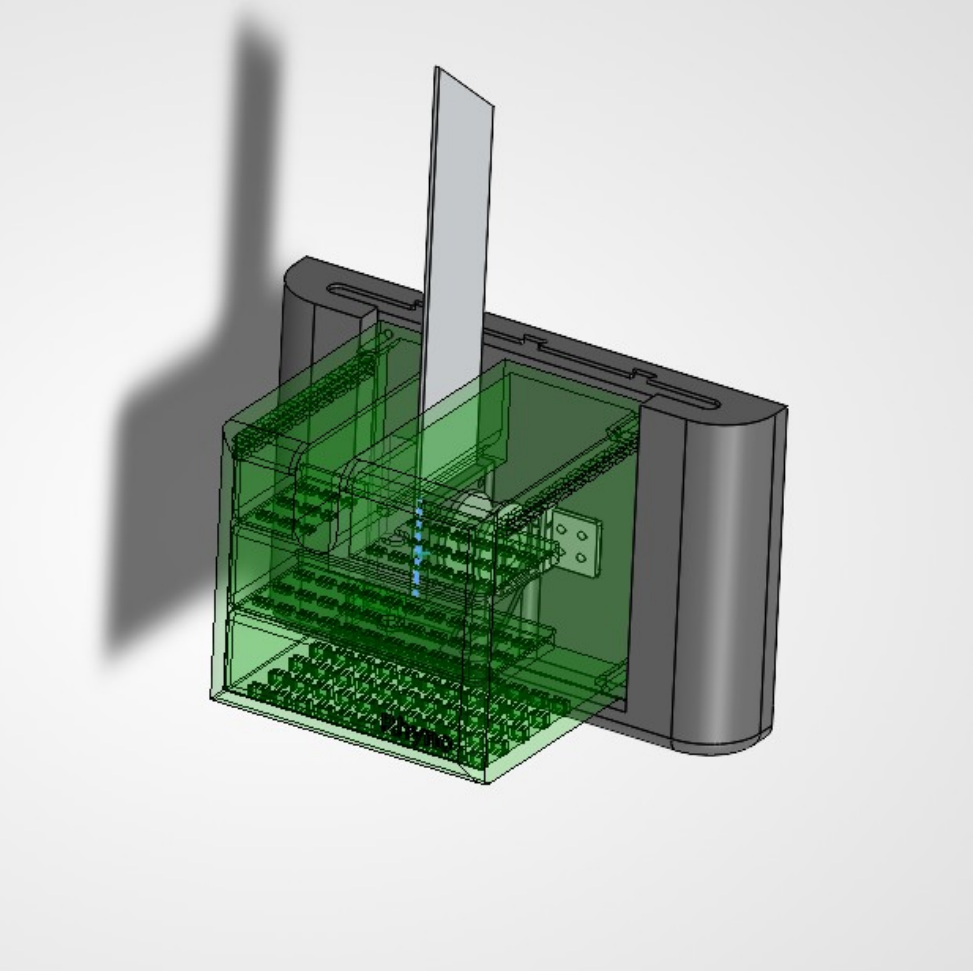
- Fingers (CAD only), then a clamp whose jaws could not close on thin cloth (Sep 30)
- v6 clamp: 4.6 times its mass budget in CAD (Oct 20)
- Final: geared four-bar, parallel jaws, 53 mm opening; gear safety factor 2.05 at servo stall
- Grip for jeans is short at narrow grabs

Grip force per jaw, jeans

$N \geq m\,(g + a) \,/\, (2\mu) = 0.9 \times 11.81 \,/\, 0.6 =$ **17.7 N**

Jaws: **13.3 N** narrowest to 25 N at a 70 mm grab

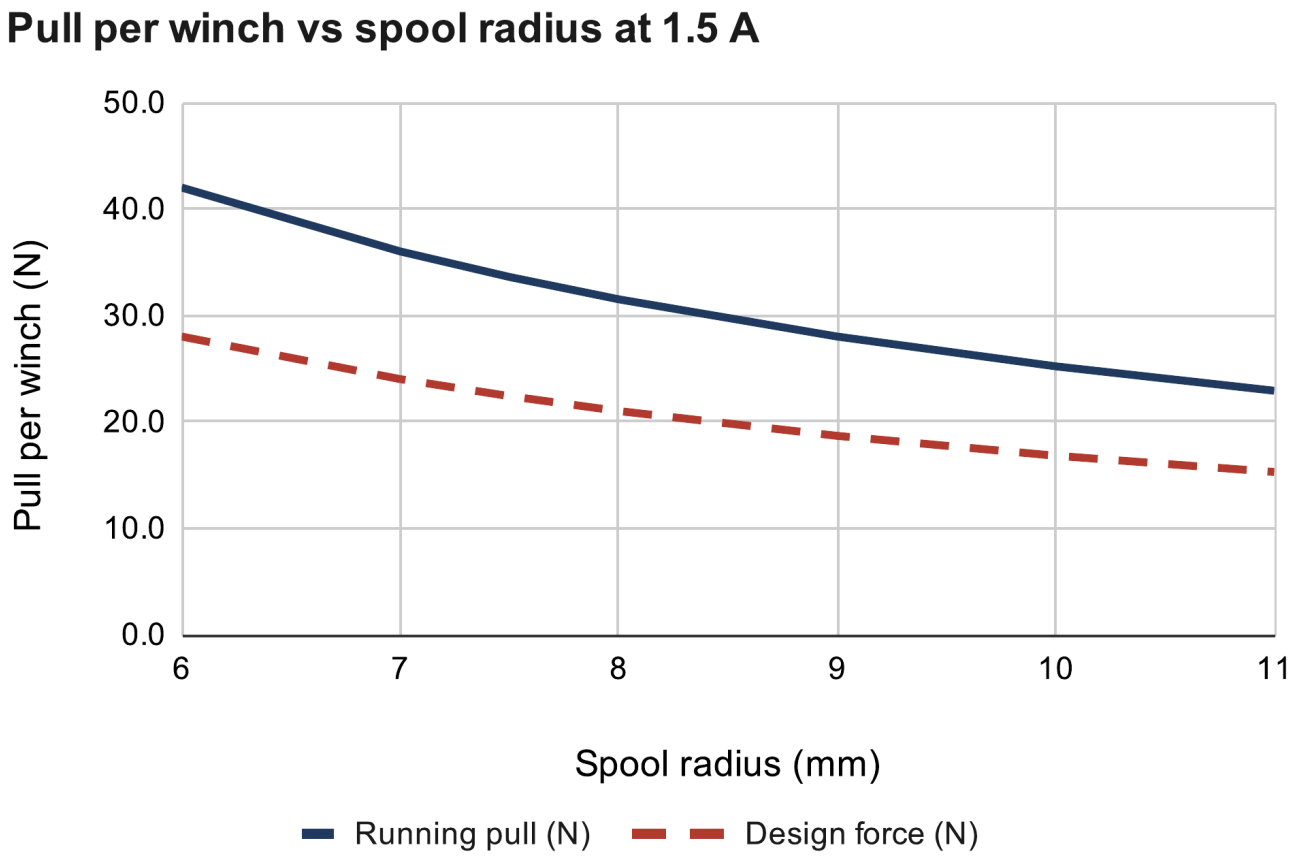
Final Wall Station



Final wall station (SolidWorks), shell shown transparent

- Slide-in wall mount: the station lifts off while the anchor stays in the stud
- Load path: cable, spool, motor, 5 mm PETG tab, shell, mount, stud
- Mount sized for a 34 N cable snag at a safety factor of 3; SolidWorks static study to run
- Aluminum fin exits through a slot in the shell top

Winch Pull at 1.5 A



Calculated (calc sheet: Winch pull)

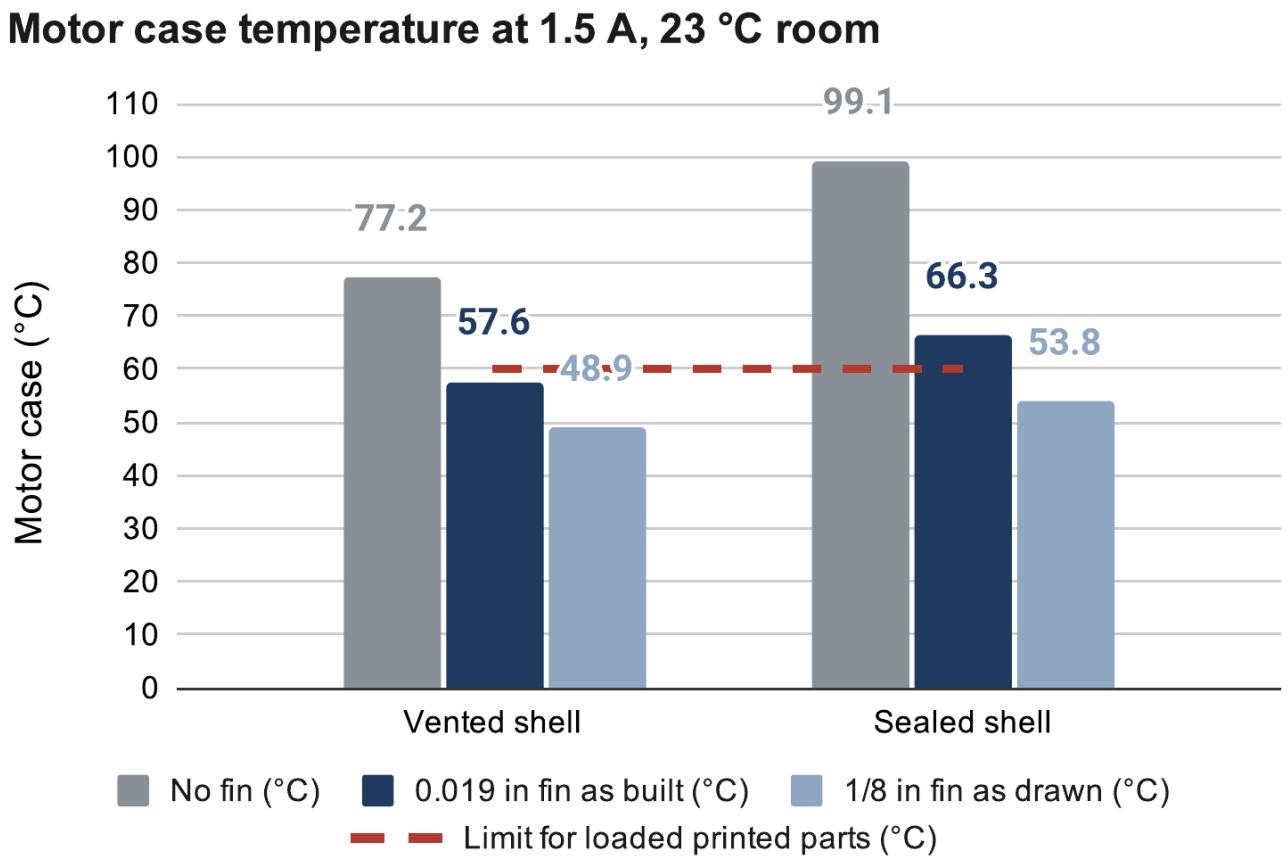
- NEMA 17 (National Electrical Manufacturers Association) stepper at its rated 1.5 A, up from 1.06 A: 41 % more pull
- Calculated pull: 34 N on an empty spool, 25 N when full
- Assumes running torque of 0.7 × holding torque; the pull test measures it

Winch design force

$T = \tau / r = 0.252\ \text{N·m} / 0.010\ \text{m} = \mathbf{25.2\ N}$

$F_{design} = 25.2\ \text{N} / 1.5 = \mathbf{16.8\ N}$

Motor Cooling



Calculated (calc sheet: Motor temperature)

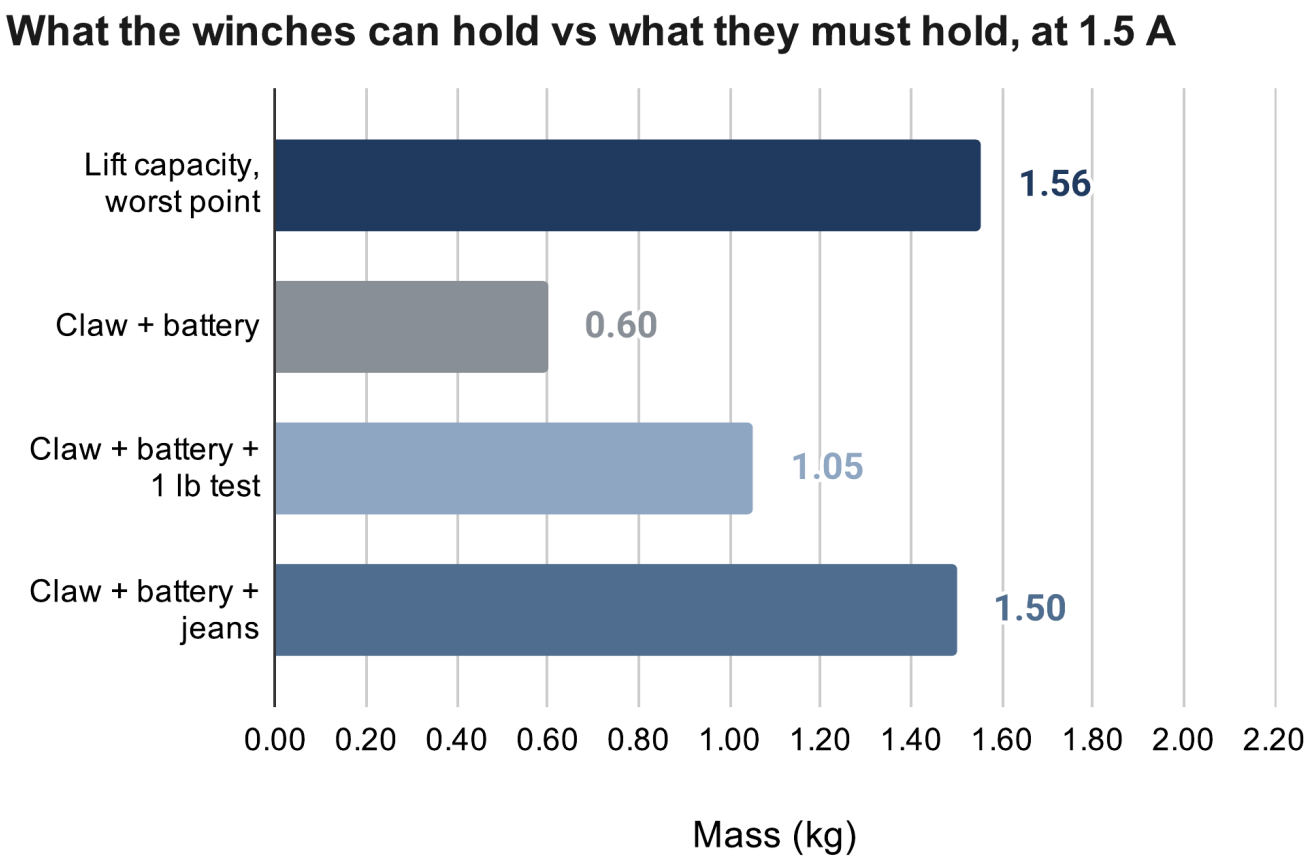
- 1.5 A doubles motor heat to 4.05 W; the PETG mount must stay below 60 °C
- Vented shell and aluminum fin: 56–59 °C with the 0.019 in prototype, 45–49 °C with the 1/8 in design fin
- Assumed: 1.8 Ω coil (to measure) and a 23 °C room; warmer air near the ceiling uses up the prototype's margin
- This week: SolidWorks study and a 90-minute hold test; fallback 1.35 A

Motor heat, $P = I^2 R$

At 1.06 A: $(1.06\ \text{A})^2 \times 1.8\ \Omega = 2.0\ \text{W}$

At 1.5 A: $(1.5\ \text{A})^2 \times 1.8\ \Omega = \mathbf{4.05\ W}$

Weight Budget



Calculated (calc sheet: Lift budget)

- Claw with battery weighed 0.60 kg (0.86 kg estimated)
- Calculated capacity at task height 1.56 kg; jeans need 1.50 kg, a 4 % margin
- That margin is inside the uncertainty: at 0.5 × holding torque, capacity is 1.1 kg
- The pull test decides: keep 1.5 A, or add a 2:1 gear stage at each winch

Lift capacity

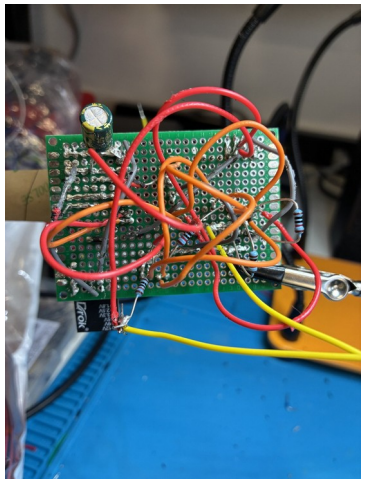
$$m_{max} = F_{design} / (k\,g) = 16.8 / (1.101 \times 9.81)$$

$$= \mathbf{1.56\ kg} \text{ against } 0.60 + 0.90 = \mathbf{1.50\ kg}$$

Build: Printing, Machining, Bring-Up









Fin blanks and drawing, finished fin, soldered station board, mill setup

- One spool bore printed too tight (Ø5.2 on a 5 mm shaft); v3 opened it to Ø5.45
- Fin drilled in one mill setup from a 1:1 drawing; the 7/8 in pilot fits the 22 mm motor boss
- First motor would not turn: logic rail at 1.14 V, traced to a bad 3.3 V connection
- Now: a meter on every rail before code, and fit coupons before new fits

First System Test

[Photo or video from the three-station test]

Check	Pass limit	Result
Self-check	All stations answer; slack taken up	*[fill in]*
Calibration	Held-out poses: cable residual ≤ *[20 mm?]*	*[fill in]*
Short move, 1 lb	300 mm at 50 mm/s; cable mismatch ≤ 2 mm	*[fill in]*
Pickup	One garment into the hamper	*[fill in]*

- Three stations hold the claw under gravity over the triangle they span, about half the floor
- Self-check, calibration, short move, then pickup
- 1 lb first, then one garment; pass limits set before the run

Results Against Targets

Target	Goal	Current	Evidence	Status
Lift capacity	1.50 kg with jeans	1.56 kg; 1.1 kg at 0.5 × holding torque	Calculated	Marginal; pull test
Grip for jeans	17.7 N per jaw	13.3–25 N	Calculated	**Short at narrow grabs**
Claw mass	0.45 kg	0.60 kg with battery	Weighed	**Not met**
Motor temperature	Below 60 °C	56–59 °C prototype fin	Calculated	Marginal; hold test
Working area	Floor, 0.2 m wall margin	All but 0.2 m at the walls	Calculated	Met in design
Wiring	One cable per station	One 12 V pair; 0.51 V drop	Built	Met
Accuracy	≤ 2 mm cable mismatch	40 ms skew budget gives 2 mm	Not measured	To be tested
Speed	50 mm/s	Not yet run	None	To be tested
Cost	*[budget]*	*[total]*	Receipts	*[fill in]*

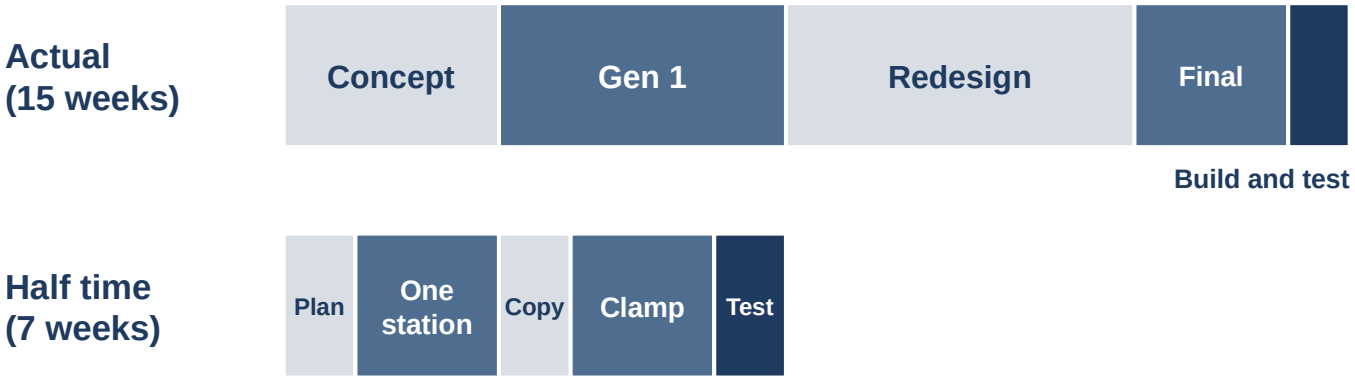
Good and Poor Decisions

Good decision	Result
One controller per winch	Five times less wire; each station tested alone
Camera calibration instead of bead homing	Whole room reachable; caught on paper before building
Teeth checked at servo stall	A clamp stalls the servo on every grip: safety factor 2.05

Poor decision	Result
Mass not tracked through four claw versions	v6 reached 4.6 times its budget in CAD; the claw is 0.15 kg over
Prototype fin cut from scrap sheet	Only 1–4 K of thermal margin
Build and test week ends on presentation day	System results arrive at the last minute

- Thorough where a mistake is expensive: the fin's single mill setup, the stall check on the teeth
- Scrappy where it is cheap to undo: a jumpered ESP32 after a controller failed, a floor board printed on paper

Takeaways and Next Steps



Same time scale. Half time: plan (pull test, mass budget), one station, copy, clamp, test.

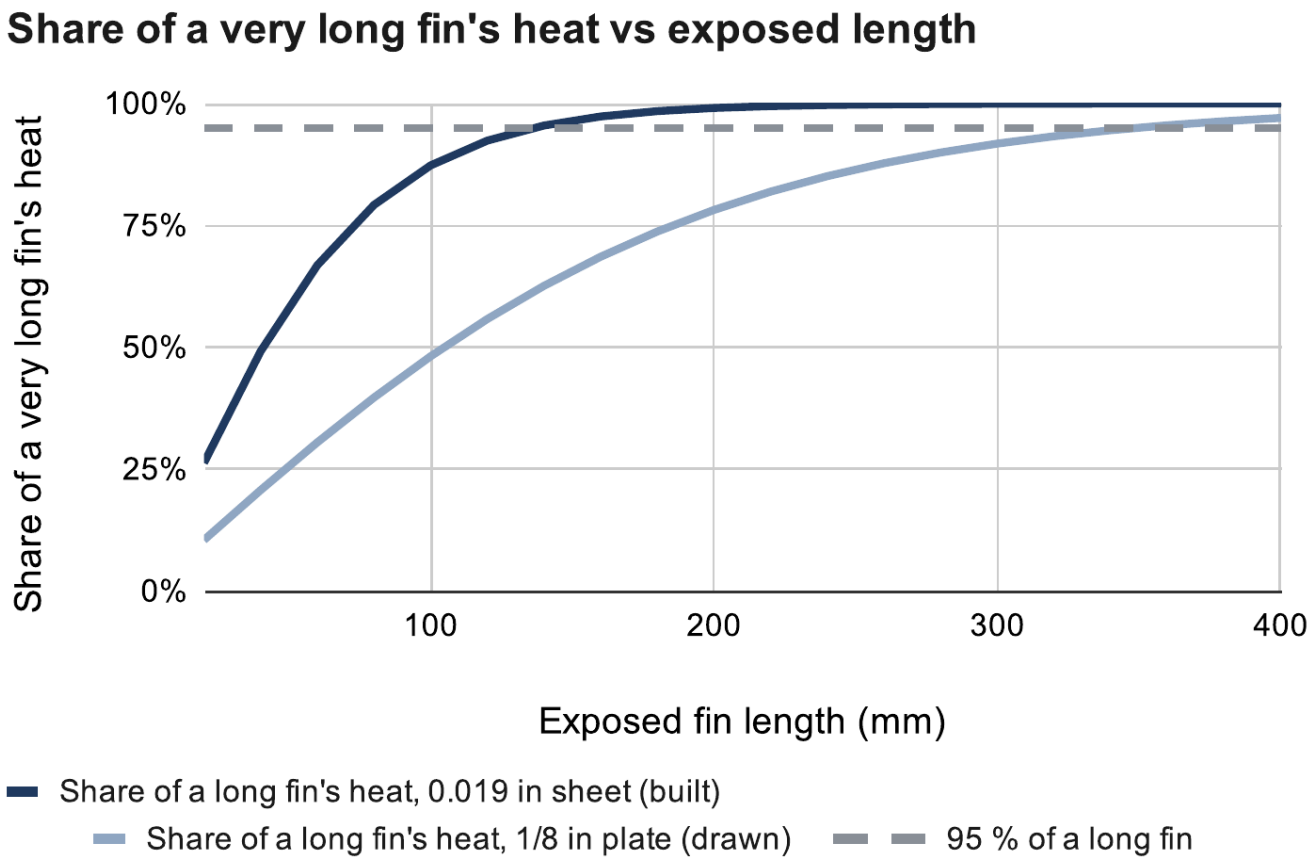
- Takeaways
  - Pull-test one winch and weigh a mock claw in week 1; set the mass budget from that
  - Kill ideas on paper first: bead homing and the central board failed calculations I could have run in week 1
  - Prove one full station, calibration included, before copying it
- Next steps
  - Pull test, then enter the measured limit in the planner
  - Hold test at 1.5 A, three-station test, grip pads
  - A brake or non-backdrivable stage, so a power loss cannot drop the claw

Appendix

Safe State and Failure Modes

Event	What happens	Mitigation
PAUSE, or a fault the software detects	Every station holds position with torque on	Nothing moves again without approval
Wi-Fi link lost	Stations stop and hold on their own clocks	Resume needs approval
Power loss or emergency stop	Torque is lost and the claw can fall (1.5 kg from about 2 m)	Nobody underneath; claw parked low when idle; brake planned
Controller reset or brown-out	Driver disabled at boot, so torque is lost	Same as power loss
Driver over-temperature	The A4988 shuts its outputs off	Hold test watches the driver; fallback 1.35 A

Heat Fin Length



Calculated (calc sheet: Heat fin)

- 0.019 in prototype reaches 95 % of its capacity by 136 mm
- 1/8 in design keeps gaining to 349 mm
- Built at 200 mm, limited by ceiling clearance

Fin length and efficiency

$m = \sqrt{(2h / (k\,t))}$, $L_{95} = 1.83 / m$

$\eta = \tanh(mL) / (mL) =$ **0.74** (1/8 in)

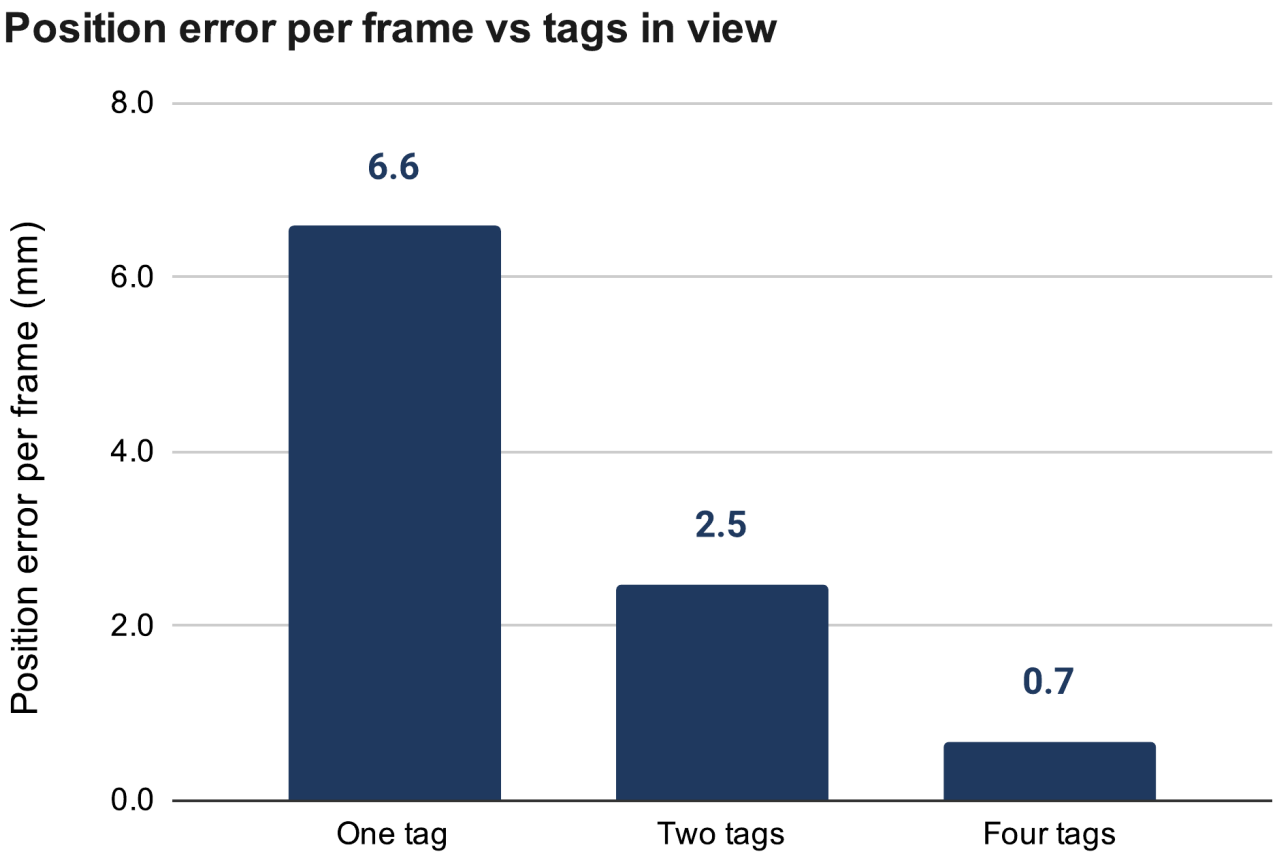
Fin Sizing for 4 W

Painted 6061, 44 mm wide	Mass	Shortest for ≤ 60 °C
0.019 in (prototype)	14 g	None (2.0 W max)
1/16 in	47 g	None (3.6 W max)
0.090 in	68 g	210 mm or none
1/8 in (design)	94 g	145–185 mm
3/16 in	141 g	125–140 mm
1/4 in	189 g	115–125 mm

All 4.05 W through the fin, 23 °C room, paste contact. Calculated.

- If the fin alone had to carry all 4.05 W
- 0.019 in sheet: not possible at any length
- 1/8 in plate: 145–185 mm

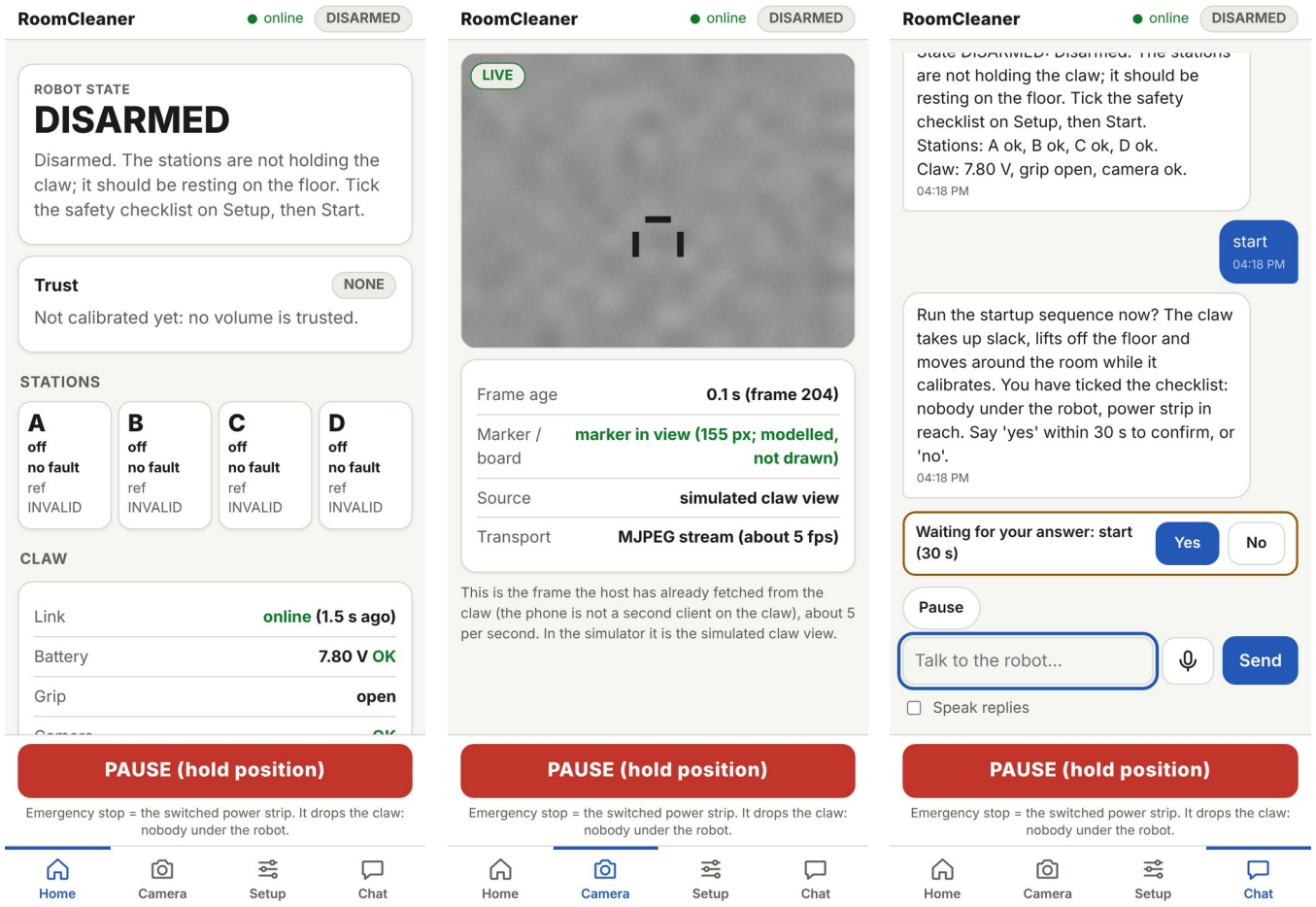
Floor Board Accuracy



- Camera pose error per frame, best case: 6.6 mm with one tag, 0.7 mm with four
- This is the camera's view, not the cable length
- After the fit, cable lengths are known to about 12–16 mm (simulated)

Calculated (calc sheet: Floor board)

Operating Sequence



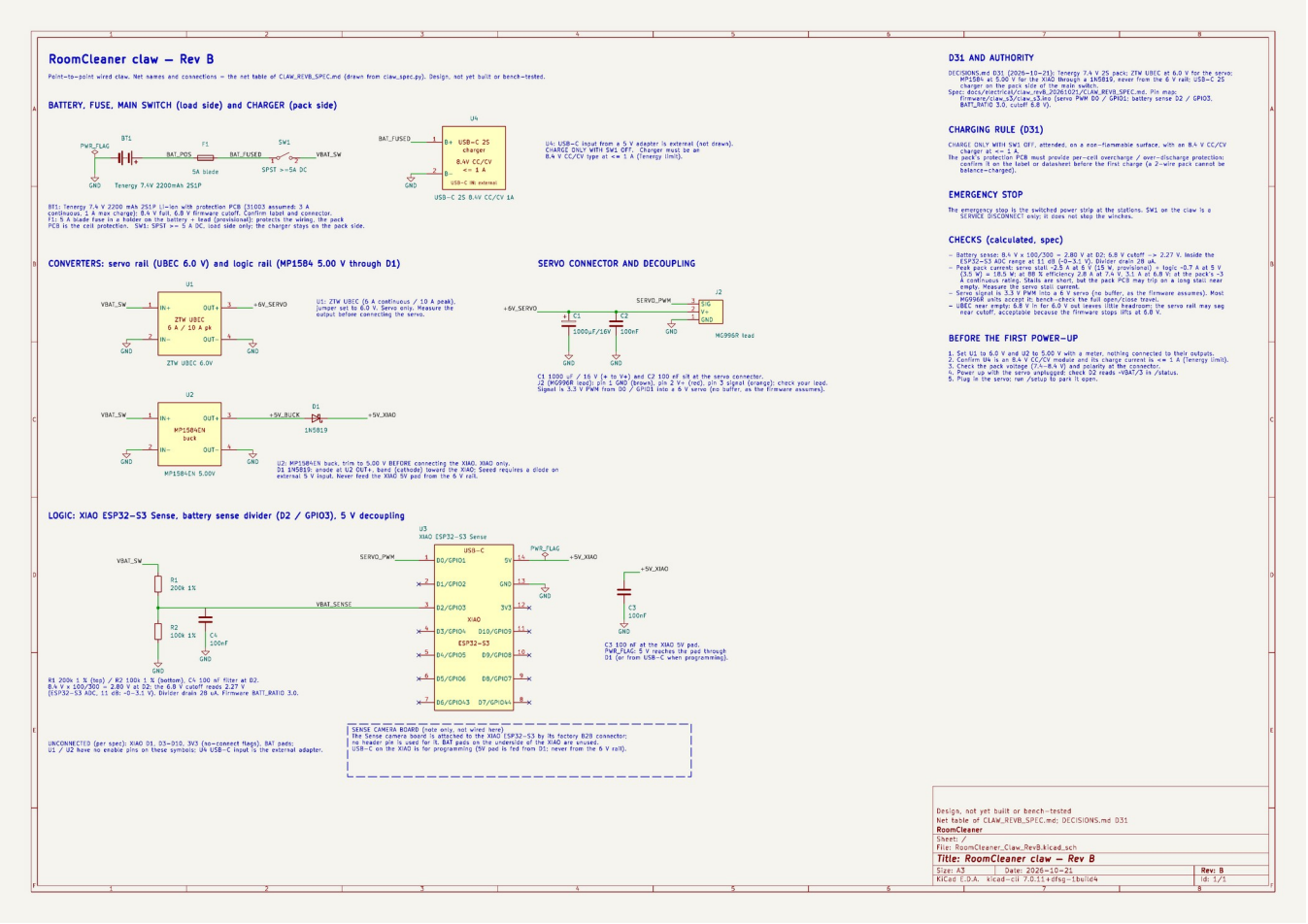
Phone app: home, live claw camera and chat

- Start up: self-check, take up slack, calibrate
- Find: scan from above, detect garments, choose the nearest
- Pick: hover until still, grip, deliver; the camera confirms each step
- Every move waits for approval on the phone; PAUSE holds position

Control Logic: Generation 1 vs. Final

	Generation 1	Final
Where is the claw?	Assumed from the commanded cable lengths	Measured by the claw camera; refit at every startup
Room geometry	Anchors from a tape measure	Anchors and cable lengths fitted, with error bars
Motor control	One central board	A controller per station, coordinated by the laptop
Finding laundry	Ceiling camera	Claw camera
Did the grab work?	Assumed	Camera checks pickup and delivery
Faults	Emergency stop only	PAUSE keeps torque; the emergency stop cuts power and drops the claw

Claw Electrical



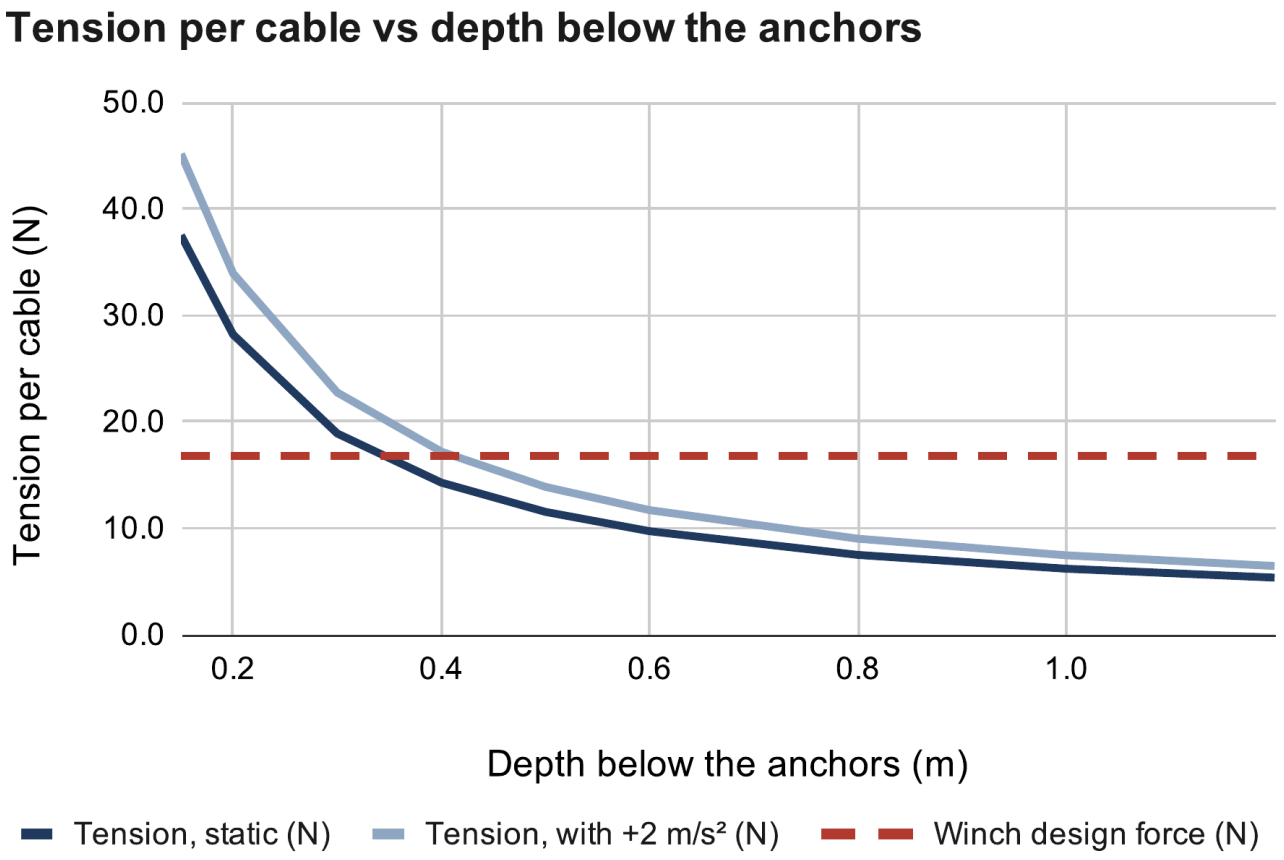
Claw schematic

- Two-cell lithium-ion pack, 7.4 V; 5 A fuse for shorts
- 6 V servo rail and a separate 5 V logic rail
- Peak draw 2.8–3.1 A, at the pack's 3 A rating

Acronyms

Term	Meaning
CAD	Computer-aided design
PETG	Polyethylene terephthalate glycol (the printed plastic)
TPU	Thermoplastic polyurethane (flexible printed plastic)
NEMA 17	National Electrical Manufacturers Association motor frame, 1.7 in square
CNC	Computer numerical control
ESP32	A Wi-Fi microcontroller family (Espressif)
A4988	Stepper-motor driver chip (Allegro)
YOLO	“You Only Look Once”: an object-detection model family
SF	Safety factor: allowable stress divided by actual stress

Calculations and Sources



Calculated (calc sheet: Cable tension)

- All charts come from my engineering calculation sheet
- Incropera et al., Fundamentals of Heat and Mass Transfer, ch. 3
- Allegro MicroSystems, A4988 datasheet

RoomCleaner engineering calculations (Google Sheets)